# Triple-Proof Quantum Finality: Post-Quantum Consensus for the Lux Network

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## Abstract

We present **triple-proof quantum finality**, a consensus mechanism for the Lux Network that achieves simultaneous classical speed, post-quantum security, and validator privacy in 248 bytes per epoch. The mechanism combines three independent cryptographic proofs: (1) BLS12-381 aggregate signatures for sub-millisecond classical consensus, (2) STARK-aggregated ML-DSA-65 (FIPS 204) signatures for post-quantum identity binding, and (3) Pulsar lattice-based threshold signatures for anonymous validator participation. No single cryptographic break compromises the system. At 1ms block times, the mechanism requires approximately 1 TB/year of storage for block headers and epoch proofs combined. We provide benchmarks showing ∼2–5 ms CPU QuasarCert verification on commodity hardware [7].

## Introduction

Post-quantum cryptography faces a fundamental tension: the most promising post-quantum signature schemes (ML-DSA, SLH-DSA) produce signatures 50–100× larger than classical schemes (ECDSA, BLS), making them impractical for per-block consensus in high-throughput blockchains. Signature aggregation, which makes BLS practical for consensus (N signatures → 48 bytes), has no known post-quantum equivalent.

We resolve this tension with a layered approach:

1. **Per-block**: Use BLS aggregation for speed (48 bytes, sub-millisecond).
2. **Per-epoch**: Use STARK proofs to compress N ML-DSA signatures into ∼200 bytes.
3. **Embedded**: Pulsar threshold signatures provide privacy within the STARK circuit at zero additional cost.

The result is a consensus mechanism that is simultaneously fast (1ms blocks), compact (248 bytes/epoch), post-quantum secure (ML-DSA + Pulsar), and privacy-preserving (anonymous validator participation).

## Threat Model

We consider three classes of adversary:

- **Classical adversary**: Can corrupt up to $f < N/3$ validators. Cannot break BLS, ML-DSA, or Pulsar.
- **Quantum adversary**: Has access to a cryptographically relevant quantum computer (CRQC). Can break BLS and ECDSA. Cannot break ML-DSA (NIST FIPS 204) or Pulsar (Module-LWE).

- **Future adversary**: One of ML-DSA or Pulsar has an unforeseen mathematical weakness. The other, combined with BLS, still provides security.

Triple-proof finality is secure against all three adversary classes simultaneously.

## The Three Proofs

### Proof 1: BLS Aggregate Signature (Classical Speed)

Every validator signs each block with their BLS12-381 key. The block proposer aggregates all signatures into a single 48-byte aggregate signature. Verification is $O(1)$ regardless of validator count.

| Property | Value |
|---|---:|
| Curve | BLS12-381 |
| Security | 128-bit classical |
| Signature size | 96 bytes (individual), 48 bytes (aggregated) |
| Aggregation | $N \to 1$ |
| Verification | $O(1)$ pairing check |
| Post-quantum | No |

BLS provides the real-time consensus path. It is fast, compact, and well-understood. Its lack of post-quantum security is compensated by Proofs 2 and 3.

### Proof 2: STARK-Aggregated ML-DSA (Post-Quantum Identity)

Every validator also signs each epoch checkpoint with their ML-DSA-65 key (FIPS 204, ML-DSA). Individual ML-DSA-65 signatures are 3,309 bytes each — too large to include in block headers.

The STARK aggregation step compresses $N$ ML-DSA signatures into a single succinct proof of approximately 200 bytes:

$$\text{STARK}(\{(\text{pk}_i, \sigma_i, m)\}_{i=1}^{N}) \to \pi \quad |\pi| \approx 200 \text{ bytes}$$

The STARK proof attests: "$N$ valid ML-DSA-65 signatures over message $m$ exist, signed by public keys $\{\text{pk}_i\}$." Verification of $\pi$ is $O(\log N)$.

| Property | Value |
|---|---:|
| Signature scheme | ML-DSA-65 (FIPS 204) |
| Security | 128-bit post-quantum (NIST Level 3) |
| Individual signature | 3,309 bytes |
| Public key | 1,952 bytes |
| Aggregation | STARK proof (∼200 bytes) |
| Verification | $O(\log N)$ |
| QuasarCert verify | ∼2–5 ms CPU [7] |

If BLS is broken by a quantum computer, the STARK proof of ML-DSA signatures remains valid. The entire chain history is verifiable post-quantum.

### Proof 3: Pulsar Threshold Signature (Post-Quantum Privacy)

Pulsar is a lattice-based (Ring-LWE) threshold signature scheme with an anonymity property: the verifier can confirm that a threshold of validators signed, but cannot determine *which* validators participated.

Pulsar operates in two rounds:

1. **Round 1**: Each participating validator generates a commitment share.
2. **Round 2**: Shares are combined into a threshold signature. The anonymity set is the full validator committee.

The Pulsar signature is embedded within the STARK circuit — the STARK proof simultaneously validates both ML-DSA identity signatures and Pulsar threshold signatures, adding zero bytes to the epoch proof.

| Property | Value |
|---|---:|
| Basis | Ring-LWE (lattice) |
| Security | 128-bit post-quantum |
| Threshold | $t$-of-$N$ |
| Anonymity | Yes (ring signature property) |
| Rounds | 2 |
| Additional bytes | 0 (embedded in STARK proof) |

Pulsar provides two properties: (1) post-quantum security independent of ML-DSA, and (2) validator privacy — no external observer can determine which specific validators signed a given epoch.

## Block Structure

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### Per-Block Header (every 1ms)

```
BlockHeader {
    parent_hash:    [32]byte    // Parent block hash
    state_root:     [32]byte    // Merkle root of state
    tx_root:        [32]byte    // Merkle root of transactions
    timestamp:      uint64      // 8 bytes
    height:         uint64      // 8 bytes
    bls_aggregate:  [32]byte    // Truncated BLS aggregate hash
    // Total: 144 bytes
}
```

The `bls_aggregate` field contains a 32-byte hash of the full 48-byte BLS aggregate signature. The full signature is recoverable from any validator's archive.

### Per-Epoch Proof (every 1,000 blocks = 1 second)

```
EpochProof {
    bls_aggregate:  [48]byte    // Full BLS aggregate signature
    zk_proof:       [~200]byte  // STARK(ML-DSA sigs + Pulsar threshold)
    validators:     uint16      // Count of participating validators
    epoch:          uint64      // Epoch number
```

```
    // Total: ~258 bytes
}
```

## Security Analysis

### Defense in Depth

The triple-proof design rests on three independent mathematical assumptions:

1. **Discrete logarithm** (BLS): Hardness of the elliptic curve discrete logarithm problem on BLS12-381.
2. **Lattice problems** (ML-DSA + Pulsar): Hardness of Module-LWE and Ring-LWE.
3. **Hash functions** (STARK): Collision resistance of the hash function used in the STARK proof system.

Any two of these three assumptions suffice for full security:

| Scenario | BLS | ML-DSA/RT | STARK |
|---|---|---|---|
| Classical adversary | ✓ | ✓ | ✓ |
| Quantum breaks DL | × | ✓ | ✓ |
| Lattice weakness found | ✓ | × | ✓ |
| STARK hash broken | ✓ | ✓(individual sigs) | × |
| Quantum + lattice | × | × | ✓* |

*If both discrete log and lattice problems are broken, only the STARK proof (hash-based) remains. This provides integrity but not authenticity — a scenario considered extremely unlikely given current understanding of quantum computing and lattice cryptography.

### Quantum Attack Window

A quantum adversary attempting to forge a BLS signature must do so within the epoch window (1 second). Current estimates for breaking BLS12-381 with a CRQC require $> 10^6$ logical qubits operating for minutes to hours. The 1-second epoch window is physically impossible to exploit with foreseeable quantum technology.

Even if the epoch window were exploitable, the ML-DSA + Pulsar proofs would detect the forgery at the next epoch boundary.

## Performance

### Benchmarks

Measured on Apple M1 Max (10 CPU cores, darwin/arm64, Go 1.26.1). GPU numbers are estimated; all others are `go test -bench` output [7].

| Operation | Time |
|---|---:|
| BLS sign | 350 $\mu$s |
| BLS verify (single) | 820 $\mu$s |
| BLS aggregated verify ($N$=100) | 875 $\mu$s |
| ML-DSA-65 sign | 495 $\mu$s |
| ML-DSA-65 verify (raw) | 181 $\mu$s |
| ML-DSA-65 verify (via fx) | 254 $\mu$s |
| ML-DSA-65 verify (cached) | 3 $\mu$s |
| Groth16 prover (CPU) | $\sim$400 ms |
| Groth16 prover (GPU, est.) | $\sim$5–15 ms |
| Groth16 verify (3 pairings, CPU) | $\sim$1–3 ms |
| Quasar block processing (BLS+ML-DSA+Pulsar) | 1.85 ms |
| **QuasarCert total CPU verify** | **$\sim$2–5 ms** |

### Storage

At 1ms block times (1,000 blocks/second):

| Component | Size | Frequency | Annual |
|---|---:|---:|---:|
| Block header | 144 B | 1,000/s | 4.5 TB |
| Epoch proof | 258 B | 1/s | 8.1 GB |
| **Total (without tx data)** | | | **$\sim$4.5 TB** |

With pruning (retain only epoch proofs after 30 days):

| Component | Retained | Annual |
|---|---:|---:|
| Recent blocks (30 days) | 144 B × 1000/s × 30d | 373 GB |
| Epoch proofs (permanent) | 258 B × 1/s × 365d | 8.1 GB |
| **Total (pruned)** | | **$\sim$381 GB** |

## Application Chains

Any chain running on the Lux primary network inherits triple-proof quantum finality. The Lux validator set provides consensus for all chains simultaneously.

Current application chains:

| Chain | Operator | Purpose | VM |
|---|---|---|---|
| Application Chain A | Partner | Regulated securities (ATS) | EVM + precompiles |
| Application Chain B | Partner | On-chain orderbook | Custom CLOB |
| Application Chain C | Partner | Encrypted computation | Threshold FHE |
| Application Chain D | Partner | AI/DeSci research | Custom |

All chains benefit from the same 248-byte quantum epoch proofs without any chain-specific implementation. Quantum finality is a property of the Lux consensus layer, not the application layer.

## Validator Key Set

Each Lux validator maintains four cryptographic keys:

| Key Type | Crypto | Purpose | NIST Standard |
|---|---|---|---|
| EC | secp256k1 (ECDSA) | EVM transactions, addresses | — |
| BLS | BLS12-381 | Consensus aggregation | — |
| ML-DSA | Dilithium-65 | PQ identity | FIPS 204 |
| Pulsar | Module-LWE | PQ anonymous signing | — |

The ML-DSA key is the validator's *long-term identity.* It certifies the BLS and Pulsar keys via ML-DSA signatures stored at validator registration (not per-block). If BLS is broken, the ML-DSA certificates prove which validator owned which BLS key.

## Conclusion

Triple-proof quantum finality provides defense-in-depth against classical, quantum, and future adversaries. By separating speed (BLS), identity (ML-DSA), and privacy (Pulsar) into independent proof paths unified by Groth16 aggregation, we achieve post-quantum consensus at 248 bytes per epoch — comparable to classical-only systems — while supporting 1ms block times on commodity hardware.

The mechanism is deployed on the Lux Network with 100+ validators and multiple application chains. Measured QuasarCert verification is ∼2–5 ms CPU, dominated by Groth16's 3 pairings on BLS12-381 [7].

## References

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